

ACFI315 Finance with Excel

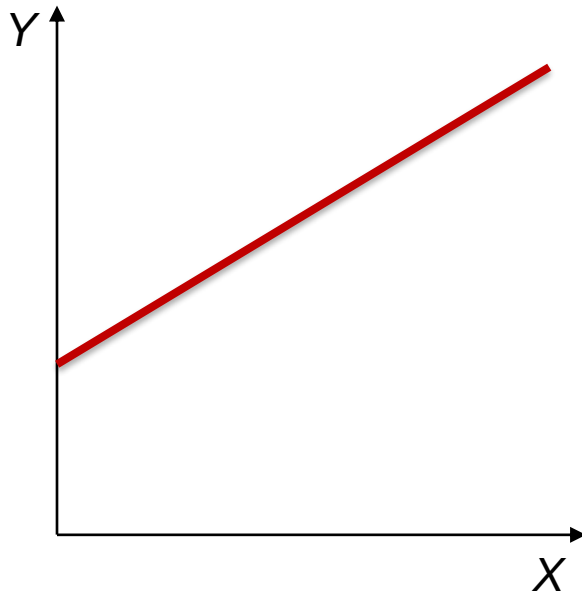
Lecture 3 Forecasting in Excel (Polynomial Relationship)



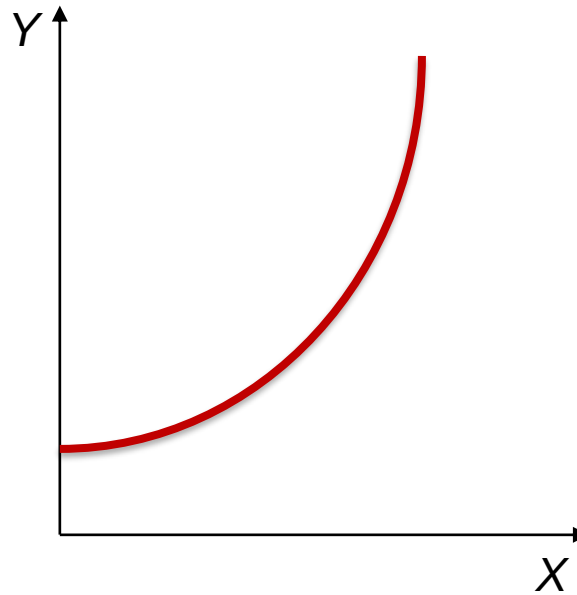
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Second Order Polynomial Relationship

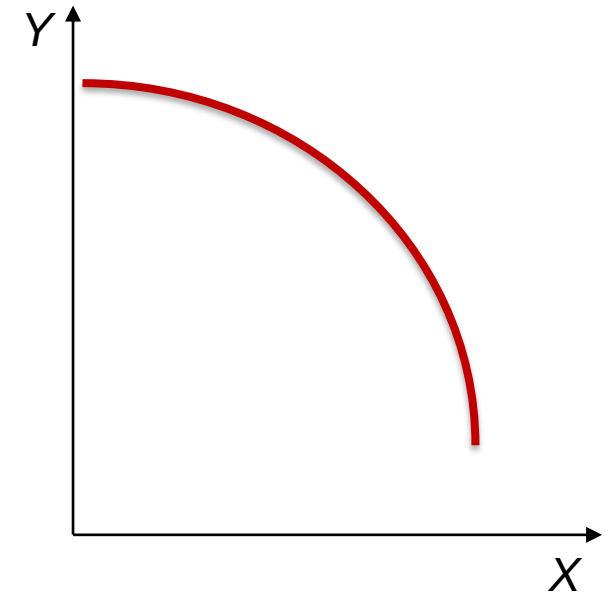
- Assume that we have the second order polynomial relationship $Y = a + bX + cX^2$ with $a > 0$ and $b > 0$, could you graph the relationship in different situations (e.g., $c = 0$, $c > 0$, and $c < 0$)?



$c = 0$



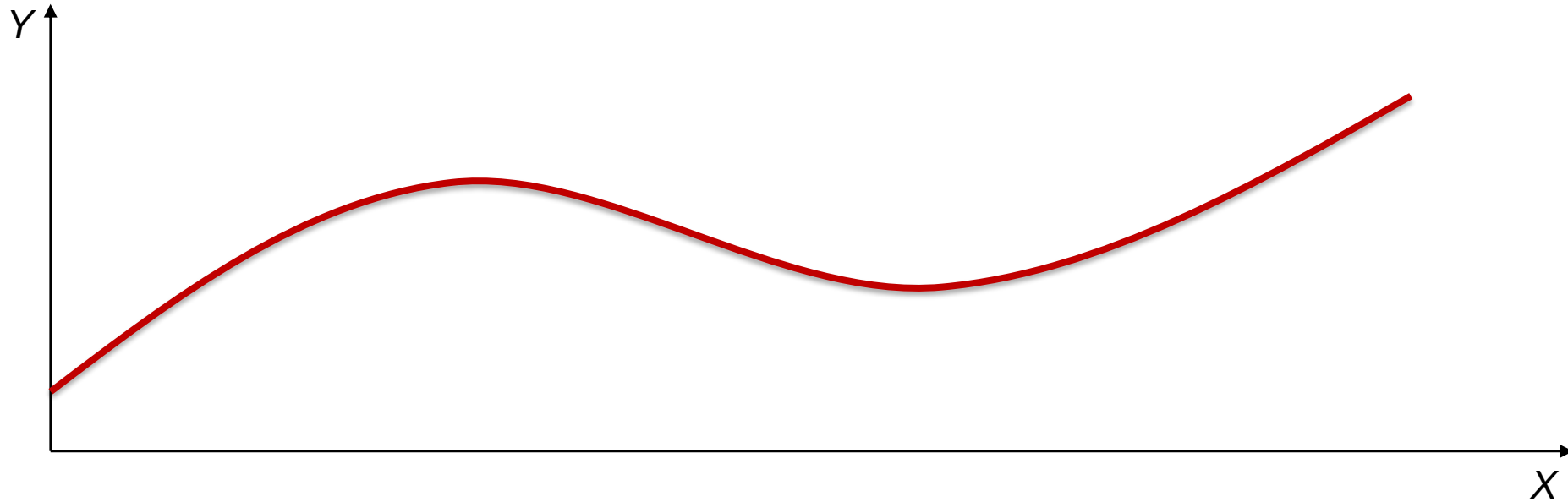
$c > 0$



$c < 0$

Third Order Polynomial Relationship

- The third order polynomial relationship $Y = a + bX + cX^2 + dX^3$ could be roughly drawn as



Linear Relationship & Polynomial Relationship

- We could use EXCEL to obtain estimates for $Y = a + bX$ and R_L^2 (R-square) for the linear model.
- We could also use EXCEL to obtain estimates for the second order polynomial relationship $Y = a + bX + cX^2$ and R_P^2 for the polynomial model.
- Imagine that $R_L^2 = R_P^2$ almost exactly, which model would you prefer. Can you explain your choice?
 - R^2 measures the proportion of variation in the dependent Y variable explained by the model. If $R^2 = 1$, the model is a perfect fit.
 - If $R_L^2 = R_P^2$ almost exactly, the linear model is explaining the same proportion of variation in the dependent Y variable explained by the polynomial model.
 - In such a situation, there would be essentially no difference between the models, especially in terms of explanatory power, which implies that $c = 0$ in the polynomial model.
 - I personally would prefer to use the linear model. Some more econometric theory could help here, but that is not massively important for our purposes on ACFI315. You could be OK to use the polynomial model, but I would be very cautious here.

Coefficients Estimation in Excel

- If we have some time-series observations of X and Y , and we want to obtain estimates of the coefficients of the second order polynomial model $Y_t = a + bX_t + cX_t^2$ in EXCEL, we could have two alternative ways.
- First, we could draw a plot of Y against X using EXCEL and then insert a trendline (In this case a first order polynomial trend. Then, we could format the trend line and ask EXCEL to display the equation and the R^2 .
- Second, we could use the **LINEST** function.
 - The **LINEST** function calculates the statistics for a line by using the "least squares" method to calculate a straight line that best fits your data, and then returns an array that describes the line.
 - We can also combine **LINEST** with other functions to calculate the statistics for other types of models that are linear in the unknown parameters, including polynomial, logarithmic, exponential, and power series.
 - Because this function returns an array of values, it must be entered as an array formula. We hit **CTRL+SHIFT+ENTER** to invoke an array function rather than **ENTER** with a scalar function.

LINEST Function

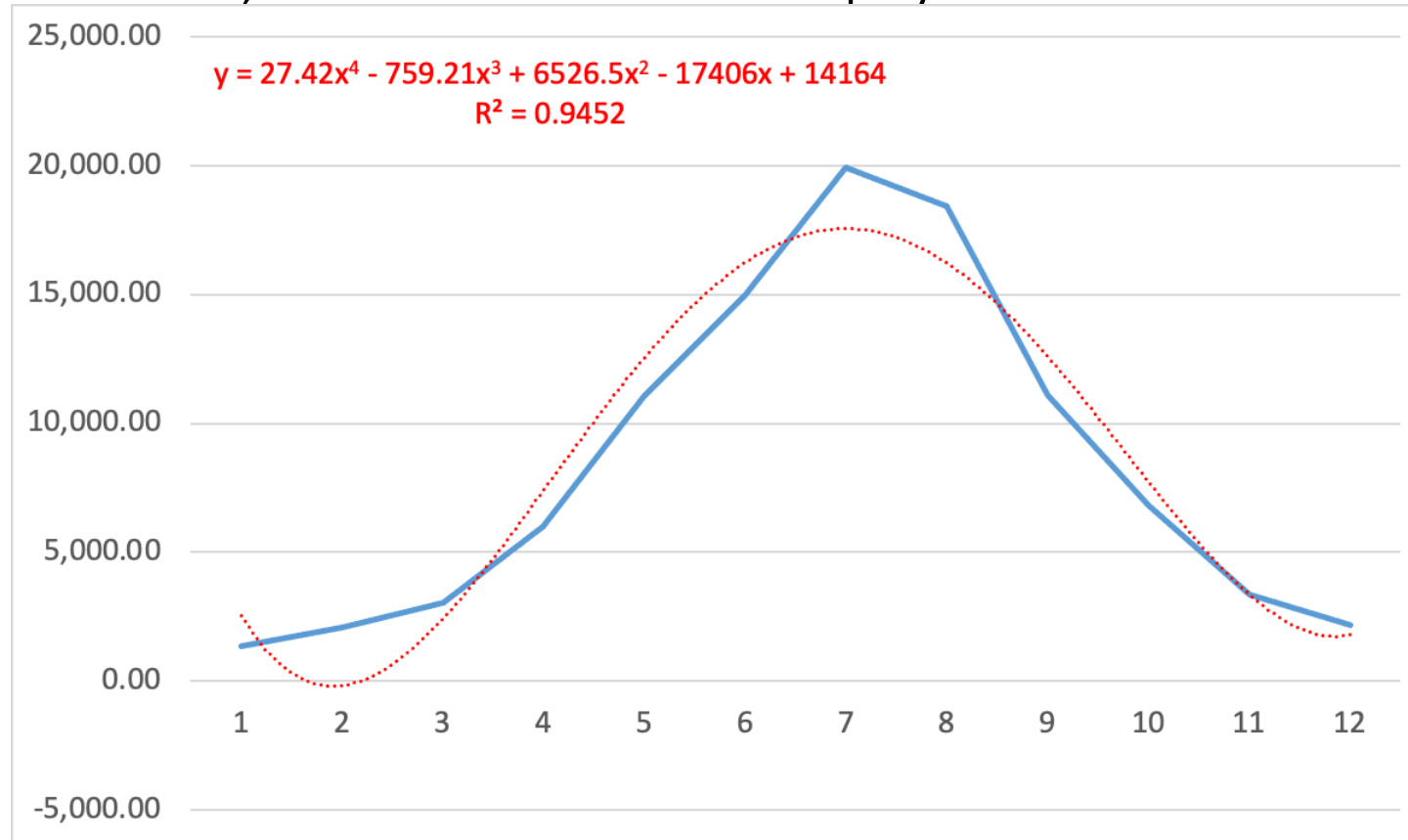
- LINEST(known_y's, [known_x's], [const], [stats])
- The **LINEST** function syntax has the following arguments:
 - **known y's** - Required. The set of y-values that you already know in the relationship $Y = a + bX$.
 - **known x's** - Optional. A set of x-values that you may already know in the relationship $Y = a + bX$.
 - **const** - Optional. A logical value specifying whether to force the constant a to equal 0.
 - **stats** - Optional. A logical value specifying whether to return additional regression statistics
- In this case, we would highlight three cells in the worksheet and then enter the following instruction:
$$=LINEST(\text{Known Y's}, \text{Known X's}^{\{1,2\}})$$
- To obtain the estimates of the coefficients of the first order polynomial model $Y_t = a + bX_t + cX_t^2$.

Forecasting in Excel

- Once we obtain coefficient estimation in EXCEL, there are two alternative ways to obtain forecasts for future periods.
- First, once we draw a plot of Y against X using EXCEL and insert a trendline, we could format the trend line and ask EXCEL to provide a forecast for future periods.
- Second, once we use LINEST function to obtain the estimates of the coefficients of the second order polynomial model $Y_t = a + bX_t + cX_t^2$, we could use these coefficient estimates $\hat{a}$, $\hat{b}$, $\hat{c}$ and future value of X to obtain future value of Y ($Y_{t+n} = \hat{a} + \hat{b}X_{t+n} + \hat{c}X_{t+n}^2$) in EXCEL.

Example

- The chart below shows the revenue of a once famous company, Blackberry, over the period from 2000 to 2012. Here, we find that a fourth order polynomial fits the data set best.



- a. Briefly explain why a linear model would not be appropriate to model this data.
- b. How well does the fourth order polynomial describe the data?
- c. How would you obtain estimates of the model using an EXCEL array function?
- d. How reliable might this estimated model be for forecasting? (Hint: Consider how much data you have available to you)

Answers

- a. Briefly explain why a linear model would not be appropriate to model this data.
 - Revenue does not increase linearly with time. Rather, revenue increases with time over the period 2000 – 2007 and then declines sharply with time following 2008.
- b. How well does the fourth order polynomial describe the data?
 - The fourth order polynomial model explains 94.52% of the variation in Blackberry revenue over the period considered. This can be interpreted as a strong description of the data.
- c. How would you obtain estimates of the model using an EXCEL array function?
 - `LINEST(known y's, known x's^{1,2,3,4})`
- d. How reliable might this estimated model be for forecasting? (Hint: Consider how much data you have available to you)
 - We are estimating five coefficients using 12 observations, so our results are always likely to be doubtful on the basis of small sample bias. Much more data would be useful here.